

Jungeun Lee

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EDUCATION

- **Ulsan National Institute of Science and Technology (UNIST)** Aug. 2020 -
Combined Master's-Doctoral Program in Electrical Engineering Ulsan, Republic of Korea
 - Focus: Multi-Agent Coordination, Reinforcement Learning, Optimization, Safe Control
 - GPA: 4.16/4.30
 - Advisor: Prof. Jeong hwan Jeon
- **Ulsan National Institute of Science and Technology (UNIST)** Mar. 2016 - Aug. 2020
B.S. in Electrical Engineering, Materials Science and Engineering Ulsan, Republic of Korea
 - Honors: *Magna Cum Laude*
 - GPA: 3.71/4.30 (Major GPA: 3.82/4.30)

EXPERIENCE

- **University of California, Los Angeles (UCLA)** Mar. 2026 -
Visiting Graduate Researcher Los Angeles, United States
 - Developed optimization and control methods for mobility systems, modeling congestion dynamics and passenger choice behavior in urban ride-sharing operations.
- **CJ Corporation** Aug. 2025 - Jan. 2026
Research Intern Seoul, Republic of Korea
 - Automated and optimized the logistics packing system using reinforcement learning.
 - Implemented a Gymnasium-based simulation reflecting real-world packing processes.
 - Reduced the total packing makespan compared to the conventional packing process through reinforcement learning and Gurobi-based mathematical optimization.

PUBLICATIONS

- [C9] **Jungeun Lee**, Kibeom Yoon, Changju Kim, Seongjae Lee, Jeong hwan Jeon. "Safe and Deadlock-Resilient Nonlinear Model Predictive Control with Bernstein Polynomials." Submitted to *65th IEEE Conference on Decision and Control (CDC)*.
- [C8] Yang Zhao, **Jungeun Lee**, Jeong hwan Jeon, Sze Zheng Yong. "Learning Safe-by-Design Neural Network Controllers." Submitted to *IEEE Control Systems Letters*.
- [C7] Yang Zhao, **Jungeun Lee**, Jeong hwan Jeon, Sze Zheng Yong. "CAffNet: Hard Constraint-Affine Neural Networks." *43th International Conference on Machine Learning (ICML)*, 2026.
- [C6] **Jungeun Lee***, Seungjae Baek*, Seongjae Lee, Sunhwi Kim, Jeong hwan Jeon. "STAR: Spatio-Temporal Rebalancing for Eco-Friendly Autonomous Mobility-on-Demand Systems." *35th International Joint Conference on Artificial Intelligence - 29th European Conference on Artificial Intelligence (IJCAI-ECAI)*, 2026.
- [C5] **Jungeun Lee***, Seongjae Lee*, Jeong hwan Jeon. "Safe Trajectory Planning with Bernstein Polynomials Under Control Barrier Function Constraints." *American Control Conference (ACC)*, 2026.
- [C4] Sunhwi Kim, Junsu Kim, Seungjae Baek, Jaechan Shin, **Jungeun Lee**, Seongjae Lee, Kyungdon Joo, Jeong hwan Jeon. "Prior-Constrained Explorative Guidance for Generalization in Diffusion Motion Planning." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [C3] Thi Thuy Ngan Duong, **Jungeun Lee**, Jeong hwan Jeon. "Cooperative Mission Planning for Heterogeneous Robots and Energy Constraints." *International Conference on Robot Intelligence Technology and Applications*, 2024.
- [J1] Cong Phat Vo, **Jungeun Lee**, Jeong hwan Jeon. "Robust Adaptive Path Tracking Control Scheme for Safe Autonomous Driving via Predicted Interval Algorithm." *IEEE Access*, 2022.
- [C2] **Jungeun Lee**, Hyeonho Noh, Hyun Jong Yang. "Performance Analysis of Beamforming for Radar and Communications Coexisting Systems." *IEEE International Conference on Information and Communication Technology Convergence (ICTC)*, 2018.
- [C1] Hyeonho Noh, **Jungeun Lee**, Hyun Jong Yang. "Beam Synthesis Under Feasible Scenarios for Radar and Communications Combined Systems." *IEEE International Conference on Information and Communication Technology Convergence (ICTC)*, 2018.

* Contributed equally

PATENTS AND SOFTWARE CERTIFICATIONS

- [S1] "Deep Reinforcement Learning and Demand Prediction Model for Mobility-on-Demand Systems." Tested and certified in accordance with KS X ISO/IEC 25023:2016 by TPR Co., Ltd. (KOLAS Accredited Testing Laboratory). Certification No. TPR-SWT-2025-017; issued on July 10, 2025.
- [P1] Hyun Jong Yang, Jonggyu Jang, Harim Lee, Yeongjun Kim, Hyeonho Noh, **Jungeun Lee**, and Hyeonsu Lyu. "Simulator for a Radar-based Active Target Detection Drone Communication Base Station." *Korean Registered Patent No. C-2018-037076*.

PROJECTS (SELECTED)

- **Real-time dynamic routing technology based on prediction and simultaneous optimization of carbon emission and transport demand** Apr. 2022 - Jun. 2025 (39 months)
Agency: Ministry of SMEs and Startups, Republic of Korea 
 - Developed a routing algorithm to solve traffic congestion and the resulting environmental pollution in metropolitan areas.
 - Designed a carbon emission model by capturing real-time vehicle sensor data on altitude, speed, and acceleration.
 - Pioneered a deep reinforcement learning-based Autonomous Mobility-on-Demand (AMoD) system to reduce carbon emissions, moving beyond the limitations of classic optimization methods.
- **AI-based applications for multi-agent mobility operations** May. 2024 - Nov. 2024 (7 months)
Agency: Electronics and Telecommunications Research Institute (ETRI)
 - Developed and modularized C++ based algorithms for task assignment optimization and optimal routing in multi-agent mobility systems.
 - Built a GUI-based simulation tool to validate algorithm performance under diverse environmental conditions and constraints.
 - Designed the system with a ROS-compatible modular architecture to enhance scalability.
- **Development of AI technology for intelligent mobility-on-demand operations** Apr. 2021 - Dec. 2022 (21 months)
Agency: Ministry of Science and ICT (MSIT), Republic of Korea 
 - Developed an optimal routing algorithm for demand-responsive mobility systems.
 - Designed the system to adjust vehicle routes in real-time, reflecting new traffic demands from nearby locations.
 - Formulated an optimization problem to minimize passenger travel delays and waiting times.
 - An average travel delay of 2.52 minutes and an average wait time of 4 minutes for ride-sharing passengers.
 - A service rate of 89.57% of the total demand.
- **AI Novatus Academia**
Agency: Gyeongsangnam-do Provincial Government, Republic of Korea 
 - Assisted in the reinforcement learning program.
 - Provided theoretical lectures on reinforcement learning for employees of small and medium-sized enterprises (SMEs) in the Gyeongsangnam-do region.
 - Conducted hands-on programming sessions to help participants apply reinforcement learning techniques to real-world applications.

SKILLS

- **Programming Languages:** Python, C++, Java, MATLAB
- **Machine Learning Frameworks:** PyTorch, TensorFlow
- **Optimization Solvers:** Gurobi, OR-Tools, IPOPT, CVX

PEER REVIEW ACTIVITIES

- IEEE Intelligent Transportation Systems Conference (ITSC) 2025, 2026
- IEEE Conference on Decision and Control (CDC) 2026

HONORS AND AWARDS

- **Human Resource Development Program for Industrial Innovation (Global)** Mar. 2026 - Aug. 2026
- **KIIE & LG CNS Optimization Grand Challenge 2025** Sep. 2025
Excellence Award (6th out of 343 Teams) 
 - Served as **team leader**.
 - Awarded \$2,200 and employment benefits.
 - Participants included researchers from leading universities (Imperial College London, Eindhoven University of Technology, SNU, KAIST, POSTECH) and major corporations (LG Display, Samsung Electronics, Hyundai Motor, KT).
- **Hyundai Motor Group Autonomous Driving Challenge** Mar. 2025
3rd Place out of 16 Qualifiers 
 - Served as **team leader**.
 - Awarded \$3,600.
 - Recognized as Korea's first end-to-end autonomous driving challenge.
 - Teams from leading universities such as KAIST, UNIST, GIST, Hanyang University, Korea University, Sungkyunkwan University.
- **Ulsan Public Data Utilization Startup Competition** Aug. 2023
3rd Place
- **National Science & Technology Scholarship** Feb. 2016 - Aug. 2020
- **Government-funded Scholarship (fully funded)** Feb. 2016 -